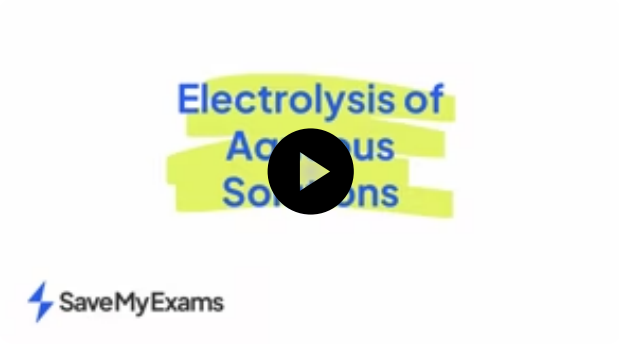
Electrolysis of Aqueous Solutions

Exam code: 0620 & 0971

Download PDF

Written by: Alexandra Brennan
Reviewed by: Stewart Hird
Updated on 8 November 2025



Did this video help you? Yes No

Electrolysis of aqueous solutions

Extended tier only

- Aqueous solutions always have water, H_2O, present
- In the electrolysis of aqueous solutions, the water molecules dissociate producing H^+ and OH^- ions:

$$H_2O \rightleftharpoons H^+ + OH^-$$

- These ions are also involved in the process and their chemistry must be considered
- We now have an electrolyte that contains ions from the compound plus ions from the water
- Which ions get discharged and at which electrode depends on the **relative reactivity** of the elements involved
- Concentrated and dilute solutions of the **same** compound give **different** products
- For anions, the **more concentrated** ion will tend to get discharged over a more dilute ion

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

Positive electrode – anode

- Negatively charged OH^- ions and non-metal ions are attracted to the positive electrode

(chlorine, bromine or iodine)

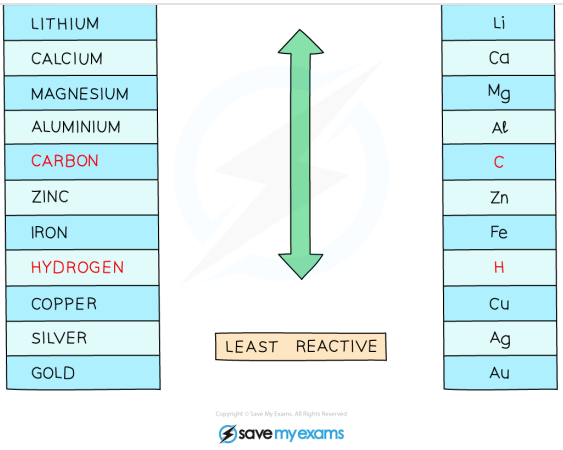
- If there are no halide ions but OH^- ions are present, oxygen is produced at the anode
  - The hydroxide ions lose electrons and forms oxygen gas (and water)
- In both cases, the other negative ion remains in solution

How concentration affects products at the anode

- The concentration of the solution affects the ion being discharged:
  - If a **concentrated** halide solution is being electrolysed, the **halogen** forms at the anode
  - If a **dilute** halide solution is being electrolysed, **oxygen** forms at the anode
- For example:
  - For concentrated barium chloride solution :
    - Cl^- ions are discharged more readily than the OH^- ions
    - So, chlorine gas is produced at the anode
  - For dilute barium chloride solution:
    - Only OH^- ions are discharged
    - So, oxygen is produced at the anode

Negative electrode – cathode

- Positively charged H^+ and metal ions are attracted to the negative electrode but only one will gain electrons
- Either hydrogen gas or metal will be produced
- If the metal is **above hydrogen** in the reactivity series:
  - The ions of the more reactive metal remain in solution
  - This causes the less reactive hydrogen ions, H^+, to be discharged
  - So, hydrogen will be produced and bubbling will be seen at the cathode
- If the metal is **below hydrogen** in the reactivity series:
  - The less reactive metal ions are discharged
  - So, the **metal** is produced and this will be seen plating onto the cathode

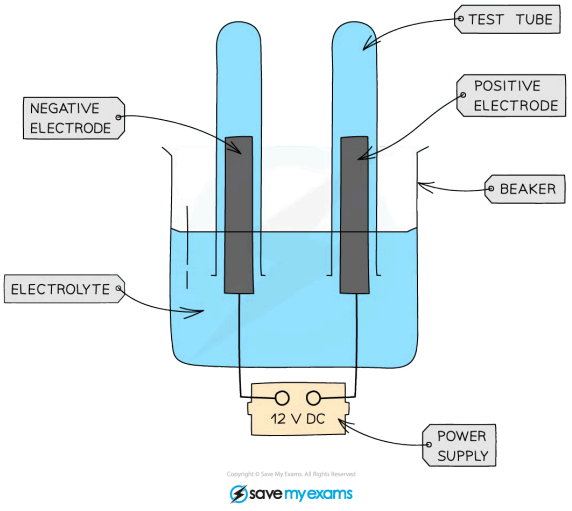


The reactivity series of metals including hydrogen and carbon

Electrolysis of aqueous copper sulfate

- Aqueous copper sulfate contains the following ions:
  - Cu^{2+} and SO_4^{2-} from the copper sulfate
  - H^+ and OH^- from the water

Using graphite electrodes:



Apparatus for the electrolysis of copper(II)sulfate using inert / passive graphite electrodes

Observations:

- The cathode is coated with a **pink-brown layer** of copper metal
- Bubbles of a **colourless gas** (oxygen) are seen forming at the anode
- The **blue colour** of the copper(II) sulfate solution **fades** over time

Product at the cathode:

- Copper ions are discharged at the cathode
  - They gain electrons and are reduced to form copper metal
- The half equation for the reaction at the electrode is:

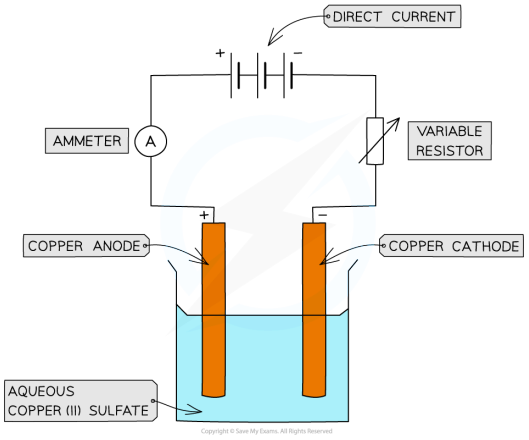
$$Cu^{2+} + 2e^- \rightarrow Cu$$

Product at the anode:

- SO_4^{2-} and OH^- are both attracted to the anode
  - OH^- ions lose electrons more readily than SO_4^{2-}
  - OH^- lose electrons and are oxidised to form oxygen gas
  - The half equation for the reaction at the anode is

$$4OH^- \rightarrow O_2 + 2H_2O + 4e^-$$

Using copper electrodes:



Apparatus for the electrolysis of copper(II)sulfate using active copper electrodes

Observations at the anode and cathode:

- The **blue colour** of the copper(II) sulfate solution **remains unchanged**
- The **cathode increases** in mass as it is coated with a layer of copper
  - This is because copper ions, Cu^{2+}, are **reduced** at the cathode and form copper atoms
- The **anode decreases** in mass as it dissolves
  - This is because copper atoms are **oxidised** at the anode and form copper ions, Cu^{2+}
- The gain in mass by the negative electrode is the **same** as the loss in mass by the positive electrode
  - Therefore, the copper deposited on the negative electrode must be the **same** copper ions that are lost from the positive electrode
- This implies that the concentration of the Cu^{2+} ions in the solution remains **constant**

Products formed for common aqueous solutions

Concentrated sodium chloride, NaCl	Chlorine gas	Hydrogen gas
Dilute sodium chloride, NaCl	Oxygen gas	Hydrogen gas
Concentrated aqueous copper(II) sulfate, $CuSO_4$	Oxygen gas	Copper
Dilute sulfuric acid, H_2SO_4	Oxygen gas	Hydrogen gas

Teacher only

New! Exclusive Teacher-Only Questions

- Fresh, expert-written questions
- Students can't access them
- Fair accurate tests - built fast

Explore in Test Builder

Test yourself

Flashcards

Was this revision note helpful? Yes No

Previous

Next

Course

Study tools

All topics

Resources

Learning Hub

Scholarship

Join

Past Papers

Dashboard

Overview

Brand ambassador

Profile

Subscription

Company

About us

Exam Specificity

Content Quality

Promotions

Jobs

Terms

Privacy

Cookie Policy

Help and Support

Subjects

Biology

Chemistry

Physics

Maths

Geography

English Literature

Psychology

All Subjects

© Copyright 2015–2026 Save My Exams Ltd. All Rights Reserved.

Neither IBO nor College Board were involved in the production of, and do not endorse, the resources created by Save My Exams.